

StemSet — Scientific Paper Template

User Manual

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Abstract

This document is the official user manual for the **StemSet** scientific paper template. It covers every feature provided by the `stemset.sty` package: paper metadata, abstract and keywords, theorem environments, coloured boxes, STEM and electrical-engineering math macros, code listings, TikZ diagram macros, algorithm pseudocode, smart cross-referencing, acronyms, SI units, and IEEE bibliography. Each feature is explained with ready-to-use examples.

Keywords: LaTeX, template, scientific paper, electronic engineering, STEM

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1 Introduction

StemSet is a professional L^AT_EX template designed for writing scientific articles (*papers*). It is optimised for electronic engineering but flexible enough for any STEM discipline.

1.1 What StemSet provides

The template bundles everything you need to produce publication-ready papers:

- **Bilingual support** — all labels switch automatically between English and Italian via a single package option.
- **Paper metadata macros** — title, authors, affiliations, corresponding email, keywords, short title for the running header.
- **Abstract and keywords** — styled and localised.
- **Theorem environments** — theorem, lemma, corollary, proposition, remark.
- **Coloured boxes** — definition, note, warning, example.
- **50+ math macros** — derivatives, transforms, phasors, linear algebra, probability, optimisation, EE-specific operators.
- **Code listings** — VHDL, C, C++, Java, Python, MATLAB with syntax highlighting.
- **TikZ diagram macros** — hardware datapaths, flowcharts, generic block diagrams.
- **Algorithm pseudocode** — via `algorithm2e`.
- **Smart cross-referencing** — via `cleveref`.
- **Acronym management** — via `acro`.
- **SI units** — via `siunitx`.
- **IEEE bibliography** — via `biblatex` + `biber`.

1.2 Language selection

To set the document language, pass the appropriate option when loading the package in `main.tex`:

```
\usepackage[english]{layout/stemset}    % English labels
\usepackage[italian]{layout/stemset}    % Italian labels
```

All localised labels (Abstract/Sommario, Keywords/Parole chiave, Theorem/Teorema, Warning/Attenzione, etc.) are set automatically.

2 Getting Started

2.1 Local installation (recommended)

The recommended setup is a local L^AT_EX installation with VS Code:

1. Install [MiKTeX](#) (Windows) or [TeX Live](#) (Linux/macOS).
2. Install [VS Code](#) and the **LaTeX Workshop** extension.
3. Extract the template folder and open it in VS Code.
4. Open `main.tex` and edit the metadata block at the top.
5. Build with the full sequence: `pdflatex` → `biber` → `pdflatex` → `pdflatex`.

Note: The Overleaf free tier may time out during compilation due to heavy packages such as `tcolorbox`, `tikz`, and `biblatex`. A local installation avoids this limitation entirely.

2.2 Overleaf

1. Go to overleaf.com and sign in.
2. **New Project** → **Upload Project** → upload the .zip file.
3. Set the compiler to **pdfLaTeX** (Menu → Compiler).
4. Compile twice for bibliography resolution.

Warning: If Overleaf times out, temporarily comment out `\usepackage{tikz}` and any TikZ diagram sections, compile, then restore them.

2.3 File structure

```
project/
+-- main.tex                <- entry point
+-- layout/
|   +-- stemset.sty <- template engine (do not edit)
|   +-- bibliografia.bib   <- bibliography
+-- sections/              <- your paper sections
|   +-- introduction.tex
|   +-- background.tex
|   +-- methodology.tex
|   +-- results.tex
|   +-- conclusion.tex
+-- immagini/              <- figures and logos
+-- .gitignore
```

Definition: Golden Rule

Only edit `main.tex` and files in `sections/`. Do not modify `stemset.sty` directly — the package is designed to be customised via the metadata macros in `main.tex`.

2.4 Two-column layout

To switch to a two-column layout (typical of IEEE conference papers), change the document class line in `main.tex`:

```
\documentclass[10pt, a4paper, twocolumn]{article}
```

2.5 Switching to Times font (IEEE standard)

Latin Modern is the default font. To switch to Times (the IEEE standard), edit `stemset.sty`:

```
% Comment out:
% \usepackage{lmodern}
% Uncomment:
\usepackage{newtxtext}
\usepackage{newtxmath}
```

3 Paper Structure

This section explains how to set up the paper metadata, abstract, and keywords.

3.1 Paper metadata

All metadata is defined in `main.tex` before `\begin{document}`:

```
\papertitle{Your Full Paper Title}
\shorttitle{Short Title} % running header
\authorlist{Author One\textsuperscript{1},
            Author Two\textsuperscript{2}}
\affiliation{%
  \textsuperscript{1}University, City, Country\\
  \textsuperscript{2}Company, City, Country%
}
\correspondingemail{you@example.com}
\paperkeywords{keyword1, keyword2, keyword3}
```

Italian aliases are available: `\titolopaper`, `\titolobreve`, `\listaautori`, `\affiliazione`, `\emailcorrispondente`, `\parolachiave`.

3.2 Title block

After `\begin{document}`, call:

```
\makepapertitle
```

This generates the formatted title block with title, authors, affiliations, corresponding email, and date. Italian alias: `\creatitolopaper`.

3.3 Abstract and keywords

```
\begin{abstract}
Your abstract text here...
\end{abstract}
\showkeywords
```

The abstract header reads “Abstract” in English or “Sommario” in Italian. The keywords header reads “Keywords” or “Parole chiave” accordingly.

`\showkeywords` prints the keywords defined by `\paperkeywords{}` and adds a horizontal rule below them.

3.4 Two-column layout

The template natively supports the IEEE-style `twocolumn` option. To use it, simply add `twocolumn` to your document class:

```
\documentclass[10pt, a4paper, twocolumn]{article}
```

The `\makepapertitle` command will automatically detect this and span the title block across both columns, while keeping the abstract in the left column.

Tip for wide elements: In a two-column layout, large tables or TikZ diagrams might overflow the column margin. To prevent this, wrap them in a `resize box`:

```
\resizebox{\columnwidth}{!}{
  \begin{tikzpicture} ... \end{tikzpicture}
}
```

For long equations, use the `split` environment from the `amsmath` package to break them across multiple lines.

3.5 Section structure

A typical paper includes:

```
\input{sections/introduction}
\input{sections/background}
\input{sections/methodology}
\input{sections/results}
\input{sections/conclusion}
```

Each file contains a `\section{}` command with a `\label{}`. Use `\subsection{}` and `\subsubsection{}` for deeper hierarchy.

3.6 Acronyms

Acronyms are managed by the `acro` package. Define them in `main.tex`:

```
\DeclareAcronym{fpga}{short=FPGA,
  long=Field Programmable Gate Array}
```

In the text, use:

- `\ac{fpga}` — first use expands to “Field Programmable Gate Array (FPGA)”, subsequent uses abbreviate to “FPGA”.
- `\acp{fpga}` — plural form.
- `\acf{fpga}` — force full form.
- `\acs{fpga}` — force short form.

3.7 SI units

The `siunitx` package provides consistent unit formatting:

```
\qty{150}{\MHz}    -> 150 MHz
\qty{4.2}{\ns}      -> 4.2 ns
\qty{3.3}{\V}       -> 3.3 V
\qty{35}{\percent}  -> 35%
```

Example 3.0 – SI units in context

A 16 bit Analog-to-Digital Converter (ADC) sampling at $f_s = 250$ MHz achieves a theoretical Signal-to-Noise Ratio (SNR) of $\text{SNR} = 6.02N + 1.76 = 98.08$ dB.

4 Math Macros

StemSet provides over 50 mathematical macros covering the most common notations in engineering and applied sciences.

4.1 Derivatives and differentials

Command	Code	Output
First derivative	<code>\der{f}{x}</code>	$\frac{df}{dx}$
Partial derivative	<code>\pder{V}{t}</code>	$\frac{\partial V}{\partial t}$
n -th derivative	<code>\nder{f}{x}{3}</code>	$\frac{d^3 f}{dx^3}$
n -th partial	<code>\npder{u}{x}{2}</code>	$\frac{\partial^2 u}{\partial x^2}$
Evaluated at	<code>\eval{f(x)}{a}{b}</code>	$[f(x)]_a^b$
Upright differential	<code>3x^2\diff x</code>	$3x^2 dx$

Example 4.0 – Integral with upright differential

The energy of a continuous signal is:

$$E = \int_{-\infty}^{+\infty} |x(t)|^2 dt$$

4.2 Transforms

Command	Code	Output
Fourier	<code>\fourier{x(t)}</code>	$\mathcal{F}\{x(t)\}$
Inverse Fourier	<code>\invfourier{X(f)}</code>	$\mathcal{F}^{-1}\{X(f)\}$
Laplace	<code>\laplace{x(t)}</code>	$\mathcal{L}\{x(t)\}$
Inverse Laplace	<code>\invlaplace{X(s)}</code>	$\mathcal{L}^{-1}\{X(s)\}$
Z-transform	<code>\ztrans{x[n]}</code>	$\mathcal{Z}\{x[n]\}$
Inverse Z	<code>\invztrans{X(z)}</code>	$\mathcal{Z}^{-1}\{X(z)\}$

4.3 Phasors and complex numbers

Command	Code	Output
Phasor	<code>\fasore{V_0}{\theta}</code>	$V_0 \angle \theta$
Real part	<code>\real{Z}</code>	$\Re\{Z\}$
Imaginary	<code>\imag{Z}</code>	$\Im\{Z\}$
Conjugate	<code>\conj{z}</code>	z^*

4.4 Linear algebra

Command	Code	Output
Transpose	<code>A\transpose</code>	A^T
Hermitian	<code>A\hermitian</code>	A^H
Diagonal	<code>\diag{\lambda_i}</code>	$\text{diag}(\lambda_i)$
Trace	<code>\trace(A)</code>	$\text{tr}(A)$
Rank	<code>\rank(A)</code>	$\text{rank}(A)$
Norm	<code>\norm{x}</code>	$\ x\ $
Absolute	<code>\abs{z}</code>	$ z $
Inner product	<code>\inner{x}{y}</code>	$\langle x, y \rangle$

Example 4.1 – 3×3 matrix

The rotation matrix in 2D can be embedded in 3D as:

$$R = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4.5 Number sets

$$\backslash NN = \mathbb{N} \quad \backslash ZZ = \mathbb{Z} \quad \backslash QQ = \mathbb{Q} \quad \backslash RR = \mathbb{R} \quad \backslash CC = \mathbb{C}$$

4.6 Systems of equations

```
\system{
  x + y &= 5 \\
  2x - y &= 1
}
```

Produces: $\begin{cases} x + y = 5 \\ 2x - y = 1 \end{cases}$

4.7 Boolean logic

`\notlog{A \cdot B} \rightarrow \overline{A \cdot B}` (complement / NOT)

4.8 Probability and statistics

Command	Code	Output
Expected value	<code>\expectval{X}</code>	$\mathbb{E}[X]$
Probability	<code>\prob{A \cap B}</code>	$\Pr\{A \cap B\}$
Variance	<code>\Var{X}</code>	$\text{Var}(X)$
Covariance	<code>\Cov{X,Y}</code>	$\text{Cov}(X, Y)$
Conditional	<code>P(A \given B)</code>	$P(A B)$

4.9 Optimisation

Command	Code	Output
Argmin	<code>\argmin_x f(x)</code>	$\arg \min_x f(x)$
Argmax	<code>\argmax_{\theta} J</code>	$\arg \max_{\theta} J$
Subject to	<code>\subjto</code>	subject to
Big-O	<code>\order{n \log n}</code>	$\mathcal{O}(n \log n)$
Big-Theta	<code>\bightheta{n^2}</code>	$\Theta(n^2)$

4.10 EE-specific macros

Command	Code	Output
Decibel	<code>60\dB</code>	60 dB
dBm	<code>-10\dBm</code>	−10 dBm
SNR	<code>\snr</code>	SNR
SINR	<code>\sinr</code>	SINR
BER	<code>\ber</code>	BER
SER	<code>\ser</code>	SER
EVM	<code>\evm</code>	EVM
Noise Figure	<code>\nf</code>	NF
Sinc	<code>\sinc(x)</code>	$\text{sinc}(x)$
Convolution	<code>x \conv h</code>	$x * h$
Circ. conv.	<code>x \circonv h</code>	$x \circledast h$
Rect	<code>\rect(t/T)</code>	$\text{rect}(t/T)$
Sign	<code>\sgn(x)</code>	$\text{sgn}(x)$

Example 4.2 – EE formula

The SNR at the output of a matched filter is:

$$\text{SNR}_{\text{out}} = \frac{2E_b}{N_0}$$

where E_b is the energy per bit and $N_0/2$ is the noise power spectral density.

5 Environments: Theorems and Boxes

5.1 Theorem environments

StemSet defines five theorem-like environments, all numbered within the current section and localised automatically.

5.1.1 Theorem

```
\begin{theorem}[Parseval]
For any signal  $x(t) \in L^2(\mathbb{R})$ :
\[ \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} |X(f)|^2 df \]
\end{theorem}
```

Theorem 5.1 (Parseval). For any signal $x(t) \in L^2(\mathbb{R})$:

$$\int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} |X(f)|^2 df$$

5.1.2 Lemma

Lemma 5.2. For all $x, y \in \mathbb{R}^n$: $\|x + y\| \leq \|x\| + \|y\|$.

5.1.3 Corollary

Corollary 5.3. If $\|x\| = 0$, then $x = 0$.

5.1.4 Proposition

Proposition 5.4. A linear time-invariant system is BIBO stable if and only if its impulse response $h(t)$ is absolutely integrable: $\int_{-\infty}^{+\infty} |h(t)| dt < \infty$.

5.1.5 Remark (unnumbered)

Remark. The numbering of theorem, lemma, corollary, and proposition shares a single counter within each section (e.g., Theorem 5.1, Lemma 5.2, Corollary 5.3).

5.1.6 Proof

Use the standard proof environment from `amsthm`:

Proof. By the Cauchy–Schwarz inequality, $|\langle x, y \rangle| \leq \|x\| \|y\|$. Setting $y = x$ yields $\|x\|^2 \leq \|x\|^2$, which holds trivially. $\square$

5.2 Coloured boxes

Four styled environments are available for highlighting content.

5.2.1 Definition box

```
\definition{Nyquist Rate}{
  The minimum sampling frequency to avoid
  aliasing:  $f_s \geq 2 f_{\max}$ .
}
```

Definition: Nyquist Rate

The minimum sampling frequency required to perfectly reconstruct a band-limited signal without aliasing:
 $f_s \geq 2f_{\max}$.

Italian alias: `\definizione{Titolo}{Testo}`.

5.2.2 Note box

`\note{Your informational note here.}`

Note: The *cleveref* package automatically determines whether to print “Fig.”, “Eq.”, or “Table” based on the label type. See [Section 8](#).

Italian alias: `\nota{Testo}`.

5.2.3 Warning box

`\warning{Your warning here.}`

Warning: Do not use `\ref` for cross-references. Always use `\cref` (lowercase) or `\Cref` (capitalised) from the *cleveref* package.

Italian alias: `\attenzione{Testo}`.

5.2.4 Example box

```
\example{Optional Title}{
  Example content...
}
```

Example 5.0 – RC time constant

For $R = 10\text{ k}\Omega$ and $C = 100\text{ nF}$:

$$\tau = RC = 10 \times 10^3 \Omega \times 100 \times 10^{-9} \text{ F} = 1 \text{ ms}$$

Italian alias: `\esempio{Titolo}{Testo}`. Examples are numbered per section (e.g., Example 5.1).

6 Code Listings

StemSet provides syntax-highlighted code listings for six programming languages, each available as a block style and an inline command.

6.1 Block listings

Use the `lstlisting` environment with the appropriate `style=` option:

```
\begin{lstlisting}[style=vhdl,
  caption={Entity declaration}, label={lst:entity}]
library ieee;
use ieee.std_logic_1164.all;
entity adder is
  port (
    a, b : in  std_logic_vector(7 downto 0);
    sum  : out std_logic_vector(7 downto 0)
  );
end entity adder;
\end{lstlisting}
```

Produces:

```
1 library ieee;
2 use ieee.std_logic_1164.all;
3
4 entity adder is
5   port (
```

```

6   a, b : in  std_logic_vector(7 downto 0);
7   sum  : out std_logic_vector(7 downto 0)
8   );
9   end entity adder;

```

Listing 1: VHDL entity declaration

6.2 Available styles

Style	Language	Details
vhdl	VHDL (IEEE 1076-2008)	Custom 4-tier keyword colouring, dedicated palette
c	C	Universal palette, standard keywords
cpp	C++	Universal palette
java	Java	Universal palette
python	Python	Universal palette
matlab	MATLAB	Universal palette

6.3 Python example

```

1  import numpy as np
2
3  def fir_filter(x, coeffs):
4      """Apply FIR filter to input signal x."""
5      N = len(coeffs)
6      y = np.zeros(len(x))
7      for n in range(len(x)):
8          for k in range(N):
9              if n - k >= 0:
10                 y[n] += coeffs[k] * x[n - k]
11
12     return y
13
14 # 4-tap moving average
15 h = np.ones(4) / 4
16 output = fir_filter(signal, h)

```

Listing 2: FIR filter in Python

6.4 C example

```

1  #include <stdio.h>
2  #define N_TAPS 4
3
4  void fir_filter(const float *x, float *y,
5                  const float *h, int len) {
6      for (int n = 0; n < len; n++) {
7          y[n] = 0.0f;
8          for (int k = 0; k < N_TAPS; k++) {
9              if (n - k >= 0)
10                 y[n] += h[k] * x[n - k];
11          }
12     }
13 }

```

Listing 3: FIR filter in C

6.5 Inline code

Each language has a dedicated inline command:

Command	Example
<code>\vhdlinline{std_logic_vector}</code>	<code>std_logic_vector</code>
<code>\cinline{int main()}</code>	<code>int main()</code>
<code>\cppinline{std::vector}</code>	<code>std::vector</code>
<code>\javainline{public static}</code>	<code>public static</code>
<code>\pythoninline{def func():}</code>	<code>def func():</code>
<code>\matlabinline{zeros(1,N)}</code>	<code>zeros(1,N)</code>

Note: The VHDL style includes a compact variant (*vhdl-compact*) with no line numbers and smaller font, useful for short snippets.

7 TikZ Diagrams and Algorithms

7.1 TikZ diagram macros

StemSet provides pre-defined TikZ node macros to create hardware datapaths, flowcharts, and generic block diagrams without writing raw TikZ code.

7.1.1 Hardware datapath blocks

Macro	Description
<code>\sysblock{id}{label}{options}</code>	Blue rounded block (hardware module)
<code>\flowblock{id}{label}{options}</code>	Orange block (process / software)
<code>\decblock{id}{label}{options}</code>	Green diamond (decision)
<code>\startendblock{id}{label}{options}</code>	Red rounded block (start/end)

7.1.2 Generic block diagram macros

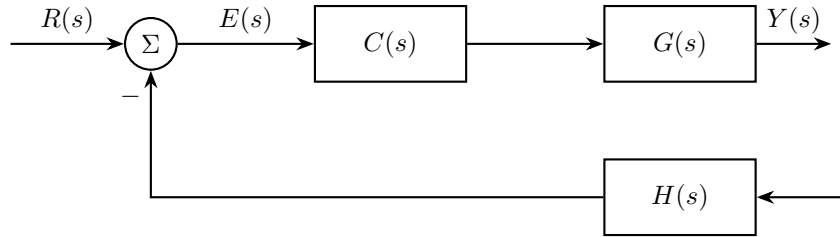
Macro	Description
<code>\genericblock{id}{label}{options}</code>	Plain rectangular block
<code>\sumnode{id}{options}</code>	Summation node (Σ)
<code>\gainblock{id}{label}{options}</code>	Triangular gain block

7.1.3 Example: signal processing chain

```
\begin{tikzpicture}[node distance=2cm,
  >=Stealth, thick]
  \sysblock{adc}{ADC}{}
  \sysblock{fir}{FIR Filter}{right=of adc}
  \sysblock{dac}{DAC}{right=of fir}
  \draw[->] (adc) -- (fir);
  \draw[->] (fir) -- (dac);
  \draw[->] ([xshift=-1.5cm]adc.west)
    -- node[above]{$x(t)$} (adc);
  \draw[->] (dac) --
    node[above]{$y(t)$} ++(1.5,0);
\end{tikzpicture}
```



7.1.4 Example: feedback control loop



7.2 Algorithm pseudocode

StemSet uses the `algorithm2e` package with `ruled`, `vlined`, and `linesnumbered` options. The algorithm label is localised (“Algorithm” / “Algoritmo”).

```

\begin{algorithm}[H]
\caption{FIR Filter Computation}
\label{alg:fir}
\KwIn{Input samples  $x[n]$ , coefficients  $h[k]$ , filter order  $N$ }
\KwOut{Filtered output  $y[n]$ }
\For{$n$ \gets 0 \to $L-1$}{
  $y[n]$ \gets 0\;
  \For{$k$ \gets 0 \to $N-1$}{
    \If{$n - k$ \geq 0}{
      $y[n]$ \gets $y[n]$ + $h[k]$ \cdot $x[$n-k$]\;
    }
  }
}
\Return{$y$}\;
\end{algorithm}

```

Algorithm 1: FIR Filter Computation

Input: Input samples $x[n]$, coefficients $h[k]$, filter order N

Output: Filtered output $y[n]$

```

1 for  $n \leftarrow 0$  to  $L - 1$  do
2    $y[n] \leftarrow 0$ ;
3   for  $k \leftarrow 0$  to  $N - 1$  do
4     if  $n - k \geq 0$  then
5        $y[n] \leftarrow y[n] + h[k] \cdot x[n - k]$ ;
6 return  $y$ ;

```

Note: Use `[H]` placement to keep the algorithm where you write it. Use `\cref{alg:fir}` to reference it: [Algorithm 1](#).

8 Bibliography and Cross-Referencing

8.1 IEEE bibliography

StemSet uses `biblatex` with the `ieee` style and `biber` backend. References are stored in `layout/bibliografia.bib`.

8.1.1 Adding references

Edit `layout/bibliografia.bib` and add entries in standard BibTeX format:

```

@article{razavi2001cmos,
  author = {Razavi, Behzad},
  title = {Design of Analog CMOS ...},

```

```

journal = {IEEE J. Solid-State Circuits},
year    = {2001},
volume  = {36},
pages   = {1050--1063},
doi     = {10.1109/JSSC.2001.XXXXXXX}
}

@inproceedings{smith2023fpga,
  author   = {Smith, John and Doe, Jane},
  title    = {Low-Power FIR Filter ...},
  booktitle = {Proc. IEEE ISCAS},
  year     = {2023},
  pages    = {1--5}
}

```

8.1.2 Citing references

Command	Output
<code>\cite{razavi2001cmos}</code>	[1]
<code>\cite{razavi2001cmos,smith2023fpga}</code>	[1], [2]
<code>\textcite{proakis2007dsp}</code>	Proakis and Manolakis [3]

8.1.3 Printing the bibliography

At the end of `main.tex`:

```

\printbibliography[heading=bibintoc,
  title={References}]
% Italian:
% title={Riferimenti Bibliografici}

```

Note: Remember to compile with the full sequence: `pdflatex` $\rightarrow$ `biber` $\rightarrow$ `pdflatex` $\rightarrow$ `pdflatex` for references to resolve correctly.

8.2 Smart cross-referencing with `cleveref`

The `cleveref` package generates context-aware references automatically.

8.2.1 Basic usage

Command	Behaviour
<code>\cref{fig:xxx}</code>	Lowercase: “fig. 1” (EN) or “fig. 1” (IT)
<code>\Cref{fig:xxx}</code>	Capitalised: “Figure 1” (EN) or “Figura 1” (IT)
<code>\cref{eq:xxx}</code>	“eq. (1)” or “eq. (1)”
<code>\cref{tab:xxx}</code>	“table 1” or “tab. 1”
<code>\cref{sec:xxx}</code>	“section 1” or “sez. 1”
<code>\cref{thm:xxx}</code>	“theorem 1” or “teorema 1”
<code>\cref{alg:xxx}</code>	“algorithm 1” or “algoritmo 1”

8.2.2 Multiple references

```
\cref{fig:a,fig:b,fig:c}
```

Produces: “figs. 1, 2 and 3” automatically.

Warning: Always place `\label{}` after `\caption{}` in figures and tables, otherwise `cleveref` may reference the wrong counter.

9 Quick Reference

This section provides a compact reference card for all StemSet commands.

9.1 Paper metadata

English command	Italian alias
<code>\papertitle{...}</code>	<code>\titolopaper{...}</code>
<code>\shorttitle{...}</code>	<code>\titolobreve{...}</code>
<code>\authorlist{...}</code>	<code>\listaautori{...}</code>
<code>\affiliation{...}</code>	<code>\affiliazione{...}</code>
<code>\correspondingemail{...}</code>	<code>\emailcorrispondente{...}</code>
<code>\paperkeywords{...}</code>	<code>\parolachiave{...}</code>
<code>\makepapertitle</code>	<code>\creatitolopaper</code>
<code>\showkeywords</code>	<code>\mostraparolachiave</code>

9.2 Coloured boxes

English command	Italian alias
<code>\definition{Title}{Text}</code>	<code>\definizione{...}{...}</code>
<code>\note{Text}</code>	<code>\nota{...}</code>
<code>\warning{Text}</code>	<code>\attenzione{...}</code>
<code>\example{Title}{Text}</code>	<code>\esempio{...}{...}</code>

9.3 Theorem environments

Environment	EN label	IT label
<code>theorem</code>	Theorem	Teorema
<code>lemma</code>	Lemma	Lemma
<code>corollary</code>	Corollary	Corollario
<code>proposition</code>	Proposition	Proposizione
<code>remark</code>	Remark	Osservazione

9.4 Math macros — compact table

Category	Command	Output	Category
Derivative	<code>\der{f}{x}</code>	$\frac{df}{dx}$	Calculus
Partial	<code>\pder{V}{t}</code>	$\frac{\partial V}{\partial t}$	Calculus
Differential	<code>\diff</code>	d	Calculus
Fourier	<code>\fourier{x}</code>	$\mathcal{F}\{x\}$	Transforms
Laplace	<code>\laplace{x}</code>	$\mathcal{L}\{x\}$	Transforms
Z-transform	<code>\ztrans{x}</code>	$\mathcal{Z}\{x\}$	Transforms
Transpose	<code>A\transpose</code>	A^T	Lin. algebra
Hermitian	<code>A\hermitian</code>	A^H	Lin. algebra
Norm	<code>\norm{x}</code>	$\ x\ $	Lin. algebra
Expected	<code>\expectval{X}</code>	$E[X]$	Probability
Probability	<code>\prob{A}</code>	$\Pr\{A\}$	Probability
Argmin	<code>\argmin</code>	$\arg \min$	Optimisation
Big-O	<code>\order{n}</code>	$\mathcal{O}(n)$	Complexity
dB	<code>60\dB</code>	60 dB	EE
SNR	<code>\snr</code>	SNR	EE
Convolution	<code>x \conv h</code>	$x * h$	EE
Sinc	<code>\sinc(x)</code>	$\text{sinc}(x)$	EE

9.5 Code listing styles

Block style	Inline command	Language
vhdl	\vhdlinline{...}	VHDL
c	\cinline{...}	C
cpp	\cppinline{...}	C++
java	\javainline{...}	Java
python	\pythoninline{...}	Python
matlab	\matlabinline{...}	MATLAB

9.6 TikZ macros

Command	Type
\sysblock{id}{label}{opts}	Hardware block (blue)
\flowblock{id}{label}{opts}	Process block (orange)
\decblock{id}{label}{opts}	Decision diamond (green)
\startendblock{id}{label}{opts}	Start/end (red)
\genericblock{id}{label}{opts}	Generic block
\sumnode{id}{opts}	Summation node
\gainblock{id}{label}{opts}	Gain triangle

References

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- [2] J. Smith and J. Doe, “Low-power FIR filter implementation on FPGA using distributed arithmetic,” in *Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS)*, 2023, pp. 1–5. DOI: [10.1109/ISCAS.2023.XXXXXXX](https://doi.org/10.1109/ISCAS.2023.XXXXXXX)
- [3] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th. Pearson Prentice Hall, 2007.